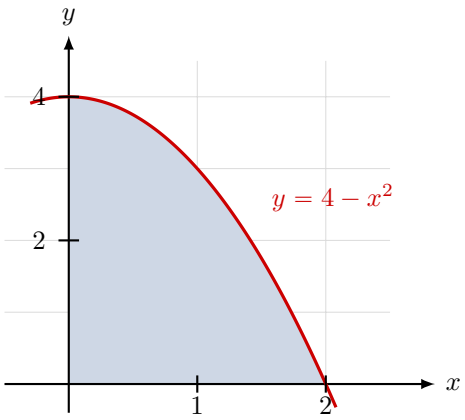
L4.3a Cylindrical Shells

Work in your group. Check your answers against the Quarto tonight.

Volume by cylindrical shells

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1. The region under $y = 4 - x^2$, $0 \leqslant x \leqslant 2$, is rotated about the y-axis.



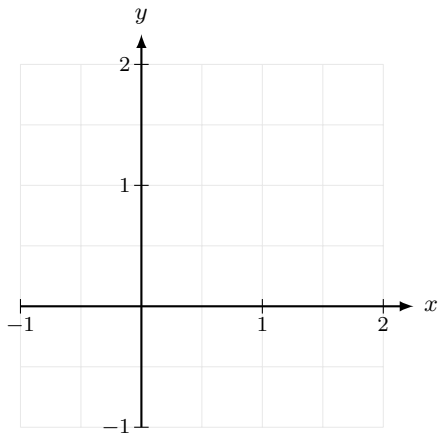
(a) Draw a representative shell on the figure. Label r and $h(r)$.

(b) Write $h(r)$, and the inner and outer r.

(c) Use shells to find the volume.

Shells for a region between two curves

2. The region between $y = 2x - x^2$ and $y = x^2$ is rotated about the y-axis.



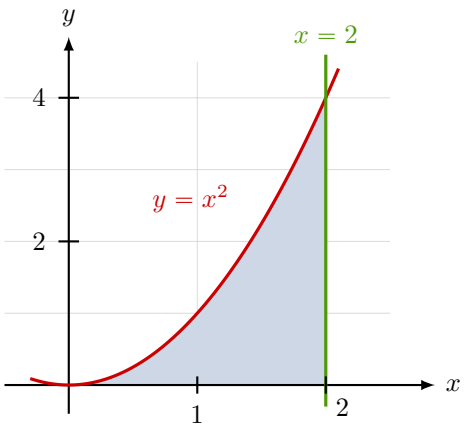
(a) Sketch the region.

(b) Test a value of x to decide which curve is on top, and write $h(r)$.

(c) Use shells to find the volume.

Shells about the x-axis

3. The region bounded by $y = x^2$, $x = 2$ and the x-axis is rotated about the x-axis.



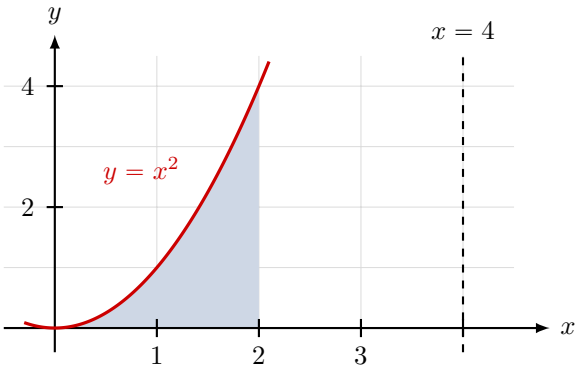
(a) Draw a representative shell on the figure. Label r and $h(r)$.

(b) Solve $y = x^2$ for x in terms of r, then write $h(r)$, and the inner and outer r.

(c) Use shells to find the volume.

Shells about a line that is not an axis

4. The region under $y = x^2$, $0 \leqslant x \leqslant 2$, is rotated about the line $x = 4$.



(a) Write r in terms of x, then x in terms of r.

(b) Write $h(r)$, and the inner and outer r.

(c) Use shells to find the volume.

5. The shaded region is rotated about the dashed line. Draw a representative shell and label r and $h(r)$. Write the volume as an integral in r. Do not evaluate.

